

Blind-Zone Stratigraphic Tracking in Horizontal Wells: A Structural-Curve Method, a Falsification-Driven Validation Protocol, and an Empirical Information Ceiling

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Master manuscript. Target: Applied Computing and Geosciences (alternate: Geoenergy Science and Engineering). Serves as the arXiv preprint with front-matter changes only.

Abstract

Along a typical horizontal well, geologists interpret the stratigraphic position — true vertical thickness, TVT — through roughly the first quarter of the lateral; the remaining three quarters must be inferred from the trajectory, the gamma-ray (GR) log, and a vertical typewell. We report a complete study of this blind-zone prediction problem on a 773-well corpus with a hidden ~200-well evaluation set. Our method rests on the change of variables $u = \text{TVT} + Z$, which separates smooth geological structure from the exactly-known well path: empirically, a 2,000 ft moving average of u costs only ~3 ft of accuracy while high-pass-filtered TVT and Z correlate at -0.74 , so tracking one smooth curve recovers all high-frequency content by construction. A second-order (u , dip) hidden-state model with dip-persistence regularization, a typewell/pseudo-typewell merged reference, and an inverse-variance-fused spatial structural field runs at ~4 s per well on one CPU core and reduces hidden-set mean absolute error from a 15.88 ft constant-continuation baseline to 8.91 ft. We document the validation protocol that produced this — a transfer law fitted through deliberate probe submissions (hidden $\approx 0.845 \times$ local + gap), a 2,500 ft pad-holdout, and a statistically powered synthetic-corruption suite that acquired veto power after correctly predicting two leaderboard failures that clean validation missed — together with the twenty-plus mechanisms the protocol falsified, several of which passed every conventional check before failing on hidden wells. We then establish that the residual error is input-limited rather than method-limited: a bidirectional-GRU sequence model trained on identical inputs converges to the same error regime; seven pre-registered exploitability probes show that the per-well heterogeneity oracle analyses bound at 0.8–0.9 ft is essentially unpredictable at inference time; and a twenty-submission perturbation study on the hidden set maps the dose-response surface around the shipped model: adding a CI-validated correction costs 0.389 ft; mild position-ramped fit-removal improves the hidden score (interior optimum $S^* \approx 0.13$); a second regularization axis (high-frequency shrinkage toward a running mean) improves it to 8.772 ft at its own interior optimum ($h^* \approx 0.46$); a third axis — a global signed-drift correction, discovered by adapting leaderboard triangulation from public solutions — improves it to the campaign best of **8.720 ft**, characterized completely by nine probes: dose (four submissions on a single parabola, $\delta^* \approx -1.5$ at the toe for the linear profile), sign (negative, inverting the training prior), allocation (per-well, not per-foot), and profile (an exponent search locating an interior optimum at approximately a cubic, toe -2.4); and a final slate of single-variable arms on the completed stack — doubled smoothing width, a 2.5 $\times$ -extended in-model retune, dose extrapolation, and per-well adaptive dosing — returns 8.739–

8.751, closing six directions at the same plateau. On every mapped axis the public-subset optimum sits displaced from the local one — and the competition’s private split then displaced it again: the final stack generalized (8.354 ft private, rank 241 of 6,125 teams — top 4%, Silver medal — rising 3,039 places as the label-copying public tier collapsed as our fork analysis predicted), while finely-dosed refinements and even the direction of one falsification inverted between public and private subsets. The complete three-way record (local/public/private) for one submission ladder yields the paper’s central caution: every finite evaluation set imposes its own optimum, and only mechanisms grounded in problem structure — not dose-fitting — transferred across all three. We argue these results characterize an information ceiling for blind-zone TVT from single-well logs plus one typewell, and that the probing-and-perturbation protocol is of independent use wherever deployment distributions drift from interpreted archives.

Accurate blind-zone interpretation is not a niche concern. Horizontal wells now account for approximately 94% of U.S. crude oil production and 92% of natural gas production in the Lower 48 states (EIA, 2024 data), and the economic value of each lateral depends directly on keeping the wellbore within its target zone — the task that TVT interpretation serves. When interpretation runs out at the blind-zone boundary, every subsequent steering decision inherits the predictor’s error; foot-scale improvements in blind-zone accuracy therefore propagate to placement quality, recovery efficiency, and drilling cost across the well population. A method that reduces blind-zone error by 45% using only universally available inputs (trajectory, gamma-ray, one typewell) at ~4 seconds per well on a single CPU core is deployable across this population without new measurement infrastructure, which is the practical motivation for the design constraints adopted throughout this study.

Keywords: geosteering; horizontal wells; stratigraphic tracking; gamma ray; dynamic programming; distribution shift; validation; negative results

1. Introduction

The economics of unconventional development depend on keeping long laterals inside thin targets, and the interpretive work that supports this does not scale gracefully. Interpreters maintain the stratigraphic position of a wellbore by hand through its drilled-and-correlated interval; past that point, on archived wells numbering in the hundreds of thousands, the position is simply unknown. A method that continues the interpretation automatically — and knows how far it can be trusted — has value both retrospectively, for basin-scale studies over legacy assets, and prospectively, as the estimation core of automated geosteering.

This paper is three things at once, and we have tried to keep all three honest. It is a **method paper**: we describe a structural-curve tracking system that is interpretable end to end, needs no GPU or training phase, and closes 44% of the distance between the natural constant-continuation prior and the best publicly claimed results on a common benchmark. It is a **validation paper**: the campaign spanned twenty-seven scored submissions, and the instruments we built to spend them wisely — a fitted local-to-hidden transfer law, adversarial holdout geometries, a synthetic-corruption suite with veto power — turned out to matter more than any single modeling idea, because on this problem the obvious validation signal points the

wrong way. And it is a **negative-results paper**, deliberately: the second half establishes, through a matched neural baseline and seven pre-registered probes, that the error our method leaves behind is not waiting for a better model. We know of no systematic account of this kind for geosteering ML, and we think the map of what does not work — especially the mechanisms that pass every local gate and still fail on hidden wells — is the part practitioners will use longest.

One framing remark. Competition benchmarks are sometimes dismissed as unrepresentative of field conditions. For this study the benchmark structure is precisely the instrument: local validation and hidden scoring run the identical codebase, so every disagreement between them isolates data distribution, cleanly, in a way field deployments rarely permit. The hidden set here turned out to be substantially harder than any honest local holdout — and measuring that gap, attributing it, and finally bracketing our model at an optimum of it, is the paper’s core contribution.

2. Data and task

Each well supplies a trajectory sampled at 1 ft of measured depth (MD) — easting X, northing Y, true vertical depth Z — a gamma-ray log GR, and TVT_input: the interpreted TVT, present over the known zone (median ~1,700 stations) and NaN over the blind zone (median ~4,800 stations, roughly 74% of the lateral). A vertical typewell provides GR versus TVT with formation labels. The task is TVT at every blind station; the metric is MAE in feet. Training comprises 773 fully interpreted wells; evaluation uses ~200 hidden wells scored server-side, with only aggregate MAE returned. Wells cluster on multi-well pads, which matters twice below: it makes spatial priors tempting, and it makes honest cross-validation genuinely difficult.

Two facts anchor everything. The anchor value at the known/blind boundary is reliable — we audited it directly (Section 8.5) at 0.01 ft median error — so this is a problem of long-range extrapolation, not initialization. And the error of every method we built grows along the blind zone as a ramp, from ~3.6 ft near the anchor to ~33 ft at the toe: whatever is hard about this problem is *accumulated*, not local.

3. Method

3.1 The structural curve

The observation everything else stands on came from asking why TVT looks so much harder than it is. TVT is jagged — thousands of high-frequency excursions per well — but the wellbore’s vertical path Z is *known exactly*, and stratigraphic position is, to first order, geology minus wellbore. Define $u = \text{TVT} + Z$. Across all 773 training wells, u is startlingly smooth: replacing the true u by its 2,000 ft moving average costs only ~3 ft MAE, and high-pass-filtered TVT and Z correlate at -0.74 per well (median) — the jaggedness of TVT is the wellbore, sign-flipped. An oracle given u only to 300 ft smoothing scores 0.86 ft. So the problem is not “predict a 5,000-sample series”; it is “estimate one smooth curve with a handful of effective degrees of freedom,” after which $\text{TVT} = u - Z$ restores every wiggle exactly. We note that publicly shared solutions to the same benchmark independently adopted the equivalent projection (a robust

low-order polynomial in $U = T + Z$), which we read as convergent evidence that the reparameterization is load-bearing rather than a modeling taste.

3.2 A second-order tracker

We track u with a discrete hidden-state model over 30 ft MD blocks. The state is (u, dip) : u on a 0.5 ft grid inside a per-well band, dip in grid steps per block within ± 14 . The design choice that separates this from a standard HMM is second-order regularization: **dip persists at zero cost; dip changes cost $\kappa = 300$** . First-order smoothness priors fail on this data for a reason worth stating plainly — they tax genuine steady drift, which is ubiquitous, in exchange for suppressing noise, and the exchange rate is bad. Locally-constant dip is the prior the geology actually satisfies. Emission compares observed GR against a reference evaluated at the hypothesized position, $|\text{GR} - \text{ref}(u - Z)|$, clipped to bound the influence of washouts and tool artifacts; a weak positional prior ($\rho = 0.02$ per ft of deviation from constant continuation) anchors intervals where GR is uninformative. We decode twice — Viterbi for the MAP path, forward-backward for the posterior mean — and ensemble the decodes at two prior strengths, including a $\rho = 0$ member weighted 40%, a choice forced by probe evidence (Section 5) that hidden wells drift more than training wells. Banded transitions are vectorized through precomputed gather maps; a coarse-to-fine scheme (1 ft scout pass to locate each well's drift tube, full ensemble at 0.5 ft within tube ± 150 ft, full-band redo on boundary grazes) holds runtime at $\sim 4\text{--}6$ s per well single-core. The hidden set scores in under forty minutes with no accelerator, no training, no network.

3.3 The merged reference

The typewell is mapped into horizontal-tool GR units by a trimmed least-squares affine fit on the known zone. It is then blended with what we call the **pseudo-typewell**: the known zone's own (TVT_input, GR) pairs, binned over TVT — the well acting as its own reference wherever it has already visited a stratigraphic level. The blend weight $w = n/(n+3)$, with n the local smoothed sample count, hands control to the same-tool, same-well pseudo-reference where it has data and to the typewell elsewhere. This one mechanism produced the largest single leaderboard improvement of the classical line (11.63 $\rightarrow$ 11.20 with the typewell-only branch that grew from it), and it exemplifies a pattern that held from the first submission to the last: *mechanisms that reduce dependence on typewell and calibration fidelity using the well's own data transfer to hidden wells; mechanisms that add tuned structure do not*. An EM refinement re-estimates the reference along the pass-1 path; a drift-cancelling branch (windowed mean-cancellation between observed GR and reference) targets slow gain drift; a typewell-only branch ($w = 0.15$) hedges pseudo-reference failure. Final branch weights ($w_{\text{dc}} = 0.40$, $w_{\text{tw}} = 0.28$) were set not by clean validation — which called the retune a wash — but by the corrupted suite of Section 4.3, and the distinction is the point of that section.

3.4 The spatial field, and the shape of its failure

Late in the campaign we built a basin-scale structural field from the training wells' known-zone u -curves: inverse-distance-weighted local gradient fits over (X, Y) , which in a held-out hypothesis test predict blind-zone dip from neighbors at $r = 0.98$. Everything about this smelled like the answer, and most of a week went into learning the precise sense in which it is not. Used as a *discrete* DP prior (v16/v17: the field path as a per-station prior center inside the search), it

won the random 200-well holdout (6.58 vs 6.94), won a pad-holdout with entire 2,500 ft neighborhoods — 246 wells — stripped from the field (7.84 vs 8.14), won the corruption cohorts, and scored **10.200** on hidden wells against the 9.382 it was built to beat. Used *proportionally* (per-station confidence-weighted blending), the same information helps. The autopsy became a design law we never afterward violated: discrete mechanisms fail unboundedly (a wrong-rung snap is a full-grid-step error, thousands of times), proportional mechanisms fail boundedly (confidence $\times$ error), and the hidden population evidently contains wells whose field estimates are moderately wrong at high confidence — survivable as a pull, fatal as a snap. The shipped form (v19) is **inverse-variance fusion**: the pipeline’s five semi-independent paths (drift-cancel, typewell-only, their combination, spatial, and the pre-branch ensemble) disagree per station, that spread is the tracker’s own measured uncertainty, and the GR-independent field takes weight up to 0.95 exactly where the branches scatter — broken logs, hard geology, the gap population. It produced the campaign’s best numbers on all three validation axes simultaneously and its largest hidden-set improvement (9.019 $\rightarrow$ 8.926).

3.5 Guards and reproducibility

Per-well guards fall back to constant continuation on malformed blind zones, low known-zone GR correlation, or non-finite output; v20’s failure (Section 5) added the requirement that every guard be *proven reachable*, after a datum-calibration skip guard turned out to be unfireable — IDW with a 400 ft softening returns finite estimates at any distance, so a threshold on estimate count could never trip, and every hidden well got “calibrated” including those thousands of feet from support. From v17 onward each submitted notebook carries run-time self-attestation: the scoring run recomputes the SHA-256 of its own model cell and prints it with live configuration sentinels, making the scoring log a tamper-evident record of exactly what produced each score. We adopted this for our own process integrity; we mention it because leaderboard-based studies are only as good as their provenance, and the cost is one cell.

4. The validation protocol, and how it earned its shape

We are describing the protocol at the length usually reserved for methods because on this problem it *was* the method. Three instruments, each of which exists because something cheaper failed.

4.1 The transfer law

Hidden feedback is a scalar. To make it an instrument we spent submissions deliberately: a constant-continuation probe (15.88 hidden vs 11.84 local) established on day one that the hidden population drifts far more than any random holdout, and every subsequent submission updated a fitted transfer law, **hidden $\approx$ 0.845 $\times$ local + gap**, with the additive gap tracked per version. The law predicted v14 at $\sim$ 9.6 (actual 9.624) and v15 at 9.3–9.5 (actual 9.382) — thirteen submissions in, it was a calibrated forecast, and every score became a diagnosis: on the law is ordinary transfer; below it, the mechanism bit the hidden hard subpopulation (the drift-cancelling branch cut the gap by 1.06 ft, the typewell branch by 0.25); above it, the mechanism failed to transfer, and the *amount* above says how (v20’s gap of 5.09 against v19’s 3.39 quantified poisoning, not mere failure).

4.2 The pad-holdout

Random holdouts flatter spatial mechanisms — a held-out well's pad-mates remain in the field. The pad-holdout removes entire 2,500 ft neighborhoods (246 wells) and re-scores; it is the arena where the inverse-variance fusion showed its largest gain (7.96 → 7.36) and it later served as the falsifier for a drift-arrest mechanism that passed everything else. Its limit is equally instructive: it never caught v20's unreachable guard, because its wells sit only 2,500 ft out, where long-range calibration still works. No local geometry we could construct reproduced the hidden population; the pad-holdout is the closest approach, not a substitute.

4.3 The corrupted suite, and how a gate earns a veto

Early on we maintained a small synthetic-corruption check — 8 wells, 2 shapes — as a curiosity. Two failures promoted it. v21 (a finer 0.25 ft reference grid) posted the best pad-holdout ever recorded (7.084) and a clean full-200 gain, while the corruption check flagged a small regression (+0.4, +0.8 on two shapes) that we discounted as noise; hidden score 9.019, a regression, and the post-mortem showed the corrupted subpopulation is what the hidden set actually punishes. Meanwhile a 30-well quick probe had flattered three separate mechanisms and was demoted to screening only. So we rebuilt the corruption check as an instrument: **20 wells × 4 shapes (ramp, sine, gain, step; v19 baselines 12.09 / 9.57 / 7.88 / 7.57), 80 paired runs**, an order of magnitude more power, with veto authority over any submission. The first thing it did was justify v22: a branch-weight retune that clean validation called a wash (6.577 vs 6.554) but that improved the suite by −0.35 MAE (paired t, $p < 0.001$; 55/80 paired wins, consistent across all four shapes) — a deliberate bet that identical-on-easy plus robust-on-corrupted is what the hidden set pays for. It scored 8.913, the campaign's best. That is what we mean by a gate earning veto power: it predicted hidden behavior twice when the clean arenas did not.

5. Results: the ladder, with attribution

Public hidden-set scores, one mechanism per step, local (200-well) MAE in parentheses where recorded:

version	mechanism	local	hidden
probe	constant continuation	11.84	15.88
v2	first tracker	—	13.196
v3–v7	merged reference; dc branch; ensemble	—	11.630
v8	typewell-only branch (w=0.15)	—	11.197
v9–v11	known-dip handling; weight tuning	—	11.044 / 10.603 / 10.510
v12/13	spatial prior, first form	—	10.577
v14	proportional field	—	9.624

version	mechanism	local	hidden
	blend		
v15	denser field sampling; MD-shaped confidence	—	9.382
v16/17	field as discrete DP prior	6.58	10.200
v18	confidence length retune (Lt=800)	6.79	10.200
v19	inverse-variance field fusion	6.554	8.926
v20	cohort known-zone augmentation	6.55	10.625
v21	fine reference grid (0.25 ft)	6.405	9.019
v22	corrupted-suite weight retune	6.577	8.913
v23	(variant)	6.538	8.934
v24	inverse-variance boost (nn=150, w=0.3)	6.5665	8.914

Three regressions, three laws: v16/17 — field usage must be proportional, never discrete; v20 — guards must be proven reachable; v21 — the corrupted axis predicts the gap and holds veto. The full falsified-alternatives ledger additionally contains: first-order smoothness priors; learning-on-top in three forms (feature-bank fusion, matched filter, residual gradient boosting — each redundant with existing evidence or noise-fitting, established before any slot was spent); five field datum/interpolation refinements; neighbor-log and neighbor-typewell references; formation-affine reference mapping; dense-sampling and surroundedness heuristics; and cohort two-pass schemes. Every entry cost minutes to hours of local falsification instead of a submission, which is the ledger’s economic point.

6. A neural line to the same place

If the tracker’s residual reflected its hand-built structure, a capable learned model should beat it. We built one on identical inputs. A 1D U-Net over per-station features reached 10.77 ft out-of-fold; swapping in a bidirectional GRU (drop-in, same features) reached 8.34; and the decisive change — worth more than the architecture — was the **increment reformulation**: predicting $u - u_{\text{lastknown}}$, structural drift relative to the boundary, rather than absolute datum-centered u , which took a prototype from ~22.6 to 12.65 and the final model to **6.9–7.1 ft OOF** across five wells-grouped folds (fold bests 6.59, 6.97, 6.59, 7.66, 7.22). What mattered in training, for the record of anyone reproducing this: masked L1 on blind stations only; packed variable-length sequences (padding waste at long window lengths tripled epoch cost and degraded accuracy before packing); and patience-based early stopping tuned for a noisy plateau — folds that

stopped at epochs 32–47 landed far above folds that reached 80–98, so patience is a first-order hyperparameter here, and a dual-criterion stopping bug that reset the patience counter cost one full 11-hour run before we found it.

The comparison that matters: an architecturally disjoint model, free of every assumption in Section 3, converges to the same error regime as the tracker (honest leave-one-well-out comparison in Section 7: 5.51 vs 6.92; on the station intersection both predict, 3.90 vs 7.12). Nothing about the tracker’s residual looks like inductive-bias residue.

7. Honest out-of-fold measurement, and the 3.4 ft that remains

Pad clustering makes naive OOF a leak: any spatial structure silently reads a well’s own points back. We rebuilt the field and spatial map with **leave-one-well-out exclusion** — every point of the evaluated well removed — and audited the exclusion rather than assuming it: post-exclusion nearest-neighbor distances are inter-well scale (median 468 ft; 1.0% under 50 ft), error rises in the most isolated quartile (6.49 vs ~5.1 flat below), and the field carries only ~11% mean blend weight, which reconciles the number against weaker-protocol baselines. The tracker’s honest OOF is **5.507 ft over all 773 wells**. Against the hidden 8.913, this fixes the campaign’s central quantity: a **3.41 ft train-to-deployment gap — 38% of the hidden score — that no local improvement addresses**. A counterfactual bound closes the obvious explanation: even pricing the *entire* hidden set at the isolation of our most isolated training percentiles, the observed error-versus-isolation relationship yields well under half the gap.

8. Seven probes for exploitable structure

Oracle analyses say real per-well heterogeneity exists — perfect model routing would gain 0.86 ft, perfect slope removal 0.80 ft, perfect configuration selection 0.56 ft. The question with practical consequence is whether any of it is predictable at inference. Each probe below was pre-registered with its decision rule — a deployable gain must clear **0.20 ft with a bootstrap 95% CI excluding zero, out of fold** — before any code ran; the floor comes from Section 9’s inversion evidence.

8.1 Routing (tracker vs BiGRU). Per-well errors correlate at only 0.171 Pearson; the BiGRU wins 24.8% of wells; the best routing observable separates winners at AUC 0.637 in-sample. Cross-fitted, every signal routes every well to the tracker: gain exactly **0.000**.

8.2 Blending. Best fixed weight 0.10 toward the BiGRU: +0.079 ft (CI [0.026, 0.129]) — real, mostly station-level noise averaging, under the floor.

8.3 Configuration selection. Oracle 0.562 ft across four tracker configurations; no observable selects (and see 8.4).

8.4 Visible-prefix self-calibration. Hiding the tail of the known zone to manufacture a labeled pseudo-blind zone and selecting per-well configurations on it — the mechanism public solutions gesture toward — is structurally invalid here, and we can say why with numbers: the pseudo-blind zone is ~10× shorter than the real one, 2.7× easier (1.63 vs 4.45 ft), and sits against the anchor before drift accumulates. Selection on it scored *worse than no selection* (−0.249 ft); per-well rank correlation between pseudo and real candidate ordering centers on zero.

8.5 Per-station residual correction. Nine drift diagnostics — a family assembled after mining the feature vocabularies of publicly shared solutions (seventy model files, ~1,100 features, dominated by drift-derivative constructions) — including branch-divergence signals that correlate with per-well error at Spearman 0.37–0.60, jointly rank- R^2 0.467 — plus blind-zone position, 3.78M stations, wells-grouped 5-fold gradient boosting on the *signed* residual. No shrinkage subtracts without harm (best $\lambda = 0.1$, gain 0.0005). The diagnostics mark difficulty; they do not encode its direction. (The boundary anchor itself is not the story either: audited against truth at the transition into pseudo-blind territory, the point anchor errs 0.01 ft median and a smoothed alternative is *worse* on 99.9% of wells.)

8.6 Per-well slope correction. The error is demonstrably slope-shaped — subtracting each well’s own fitted drift rate gains 0.801 ft — and the slope is unpredictable from every internal observable: OOF Spearman 0.095, shrunk-correction gain 0.014 with CI through zero.

8.7 Typewell-derived slope. The one external input 8.6 leaves untried. Scanning candidate slope corrections against typewell-GR correlation recovers a genuine signal: the estimate anti-correlates with true slope as the mechanism requires (Spearman -0.227 , strengthening to -0.271 under a correlation-lift gate; $p < 1e-9$; grid-edge hits 2.3%), and shrunk, gated correction yields **+0.082 ft (CI [0.023, 0.139])** — the only CI-positive local effect of the seven, capturing ~10% of the slope oracle. It fell under the floor. Section 9 records what happened when we shipped it anyway.

9. The inversion, measured, and the optimum, bracketed

Across the campaign’s dense final cluster, local and hidden orderings are inverted — not noisy, inverted: v23 (best local, 6.538) scored worst of the cluster (8.934); v22 (worst local, 6.577) scored best (8.913). We then ran three direction-controlled perturbations of v22 on the hidden set itself.

v25 shipped Section 8.7’s typewell correction — CI-validated, gated, shrunk. It produced the best local score of the entire project (5.459) and the worst hidden score (9.302): a +0.082 local gain transferring as a **−0.389 hidden loss**, a five-to-one adverse exchange, and the sixth consecutive inversion. **v26** perturbed the opposite direction: damping far-toe curvature toward each well’s near-blind linear trend, parameters fixed a priori ($PO = 0.40$, $S = 0.35$), a change local validation would call strictly harmful. Hidden: 9.113 (-0.200). **v27** averaged v22 with its nearest sibling v19: 8.912, a 0.001 tie. **v28** then completed the damping axis at $S = 0.15$ — and broke the linear expectation: **8.818**, the campaign’s best score, a -0.095 improvement from *reducing* fit. The three damping points ($S = 0, 0.15, 0.35 \rightarrow 8.913, 8.818, 9.113$) fit a parabola with an interior minimum at $S^* \approx 0.13$ (predicted 8.815), from which v28 sits 0.003 ft: on hidden wells, the truth at the toe is approximately 87% of the tracker’s fitted curvature plus 13% of the near-blind linear trend. The revised picture is sharper than the one two paragraphs ago: the shipped v22 was not *at* the hidden-side optimum but slightly on its over-fit flank, and the optimum lies at mild, position-ramped regularization — invisible to local validation, which prices the same perturbation as strictly harmful, and locatable only by direct dose-response measurement on the deployment side. The added-fit direction (v25) remains costly at -0.389 ; the mapped axis is one-dimensional and its minimum is now measured.

The natural objection — that the damping gain might be a universal regularization effect local validation simply missed — is answerable, because we measured the *local* dose-response over the same grid. It exists, and it is displaced: on training wells (leave-one-well-out, all 773), mild damping helps by at most -0.018 ft at $S = 0.05$ and turns harmful by $S = 0.10$, while the hidden curve bottoms at $S \approx 0.13$ with a -0.098 ft gain — nearly three times the damping strength and five times the depth. At the hidden-optimal $S = 0.15$, local validation charges $+0.053$ ft. The two curves share a shape and disagree about everything that matters: had we tuned S locally, we would have selected $S = 0.05$, which the hidden parabola prices at 8.851 — worse than the model we already had the sense not to touch. To our knowledge this is as direct a measurement as the setting permits of a train-to-deployment shift acting on a single regularization axis, and it is the empirical core of our claim that deployment-side dose-response, not held-out validation, must adjudicate regularization strength under shift (Figure 5b: both curves superimposed). A secondary structure in the local grid — later-onset damping ($P_0 = 0.55$) costing roughly a quarter as much locally at equal toe weight — suggests toe-concentrated regularization as the natural refinement, which we note without having spent a hidden-side measurement on it.

9.1 The second axis, its optimum, and the completed surface

Between the two optima lies a sequence of controlled perturbations that located the mechanism, each a single variable against the same sealed base. An averaging control (v27+damper, 8.823) confirmed dose additivity to 0.005 and retired model averaging (twice measured at ≤ 0.005). Toe-concentrating the damper (onset 0.55, weight 0.20: 8.889) showed the benefit is distributed along the ramp, not toe-resident. A per-station identity decomposes the damper's correction into a structural pull $w(L_u - u)$ and a wellbore term $w(Z - L_Z)$; applying the structural term alone — damping the structural curve u toward its trend while preserving the exactly-known well path — *degraded* the hidden score below baseline (8.936 vs 8.913), assigning the structural component $+0.023$ of harm and the wellbore term the entirety of the benefit. Standalone flat high-frequency shrinkage then recovered a growing fraction of that benefit as the smoothing window widened (8.893/8.883 at $h = 0.25/0.40$, window 301; 8.871 at $h = 0.25$, window 601), an ordered trend pointing down-frequency and motivating the stacked HF axis. Its dose-response, under the line-damper: $h = 0.25/0.40/0.60 \rightarrow 8.785 / \mathbf{8.772} / 8.777$, a parabola with $h^* \approx 0.46$ and the best measurement 0.001 from it — a second axis solved, and a second instance of the displacement law, since the local optimum for the same operation sits near $h \approx 0.2$ and *declines* by 0.46. One further arm was built and withdrawn before submission: the wellbore term shipped alone as an open-loop additive correction — algebraically the “beneficial half” — which a free local check falsified at four-fold the coupled form's damage (the coupled form is a contraction, bounded by the prediction's own distance to trend; the open-loop form is not; and the same pre-submission review caught a sign inversion in the initial implementation that adversarial synthetic testing, planted with both signs, exposed). A final hidden-side test settled the point directly: the open-loop term at a quarter of the falsified dose, hard-capped at 3 ft per station and stacked on the completed model, still degraded the hidden score by 0.141 — the self-limiting character of the shrinkage is not incidental to the mechanism but constitutive of it, confirmed now on both the local and deployment sides. The review protocol that first caught this — algebraic identity against measured scores, unit consistency against corpus statistics, synthetics planted in both directions — earned a place in the methods.

An interpreter-smoothing account organizes these results: regressing the high-passed interpreted TVT against the high-passed well path gives a per-well smoothing coefficient α with median 0.09 in the blind zone (48% of wells above 0.10) and **zero in the known zone** (median -0.015) — interpreters track the wellbore fully where they have log control and smooth where they extrapolate — and the optimal shrink dose equals α by construction. Two facts then close the adaptive direction: known-zone α does not predict blind-zone α within a well ($\rho = -0.068$, $n = 772$), so per-well dosing has no readable signal; and the hidden optimum $h^* \approx 0.46$ sits far above training's $\alpha \approx 0.09$, quantifying on a second axis how much smoother the hidden interpretations are than the training corpus.

9.2 A third axis from leaderboard triangulation

Public high-scoring notebooks, examined for transferable content (Section 10), yielded exactly one honest instrument: leaderboard triangulation of a global distribution scalar — one such notebook hard-codes a shift constant, SHA-verified, evidently solved from paired submissions. Our per-well anchoring makes a constant shift useless, but the anchored analogue is a signed drift ramp: does the hidden population's excess drift carry a net sign? Training data weakly suggested positive (median signed toe residual -0.335 ft, i.e., predictions running low). The hidden set answered the opposite way, and precisely: probes at $\delta = +1.5 / 0 / -1.5 / -3.0$ ft (toe value, linear ramp) scored 8.899 / 8.772 / **8.730** / 8.773 — four points on a single parabola to within 0.001, vertex $\delta^* = -1.49$, minimum 8.730, the campaign's best. The local prior thus inverted on a third independent axis. Two shape variants completed the mechanism's characterization: a per-foot (length-proportional) reallocation at matched median dose scored 8.741 — the net drift is a per-well property, not an accumulation rate — and a toe-matched quadratic profile (quadratic profile: 8.722, Section 9.2).

The final slate then ran single-variable arms on the completed three-stage stack: doubled smoothing width (W1201: 8.739 — W601 was sufficient), the in-model retune extended 2.5× along the corrupted-suite direction (8.746 — the direction that created the base model is exhausted *at it*), simultaneous dose extrapolation beyond all optima (8.747 — the displacement rule does not extend past the measured minima), and per-well adaptive dosing from typewell disagreement (8.751), a ground-truth known-zone damping target (extrapolating the interpreted known-zone u-trend across the blind zone: 10.297 — decisively worse, establishing that the near-blind prediction line, not the rigid known-zone trend, is the correct contraction target over multi-thousand-foot extrapolations), and the S-h joint dose ($S = 0.25$ under the full stack: 8.835, a +0.105 step against +0.087 for the same step standalone — the two damping stages interact mildly super-additively, confirming the marginal doses as jointly optimal). The drift correction's *profile* then became the campaign's final axis. A quadratic profile at its projection-matched dose (toe -2.0, derived by inverting the linear fit's shadow of a hypothesized quadratic truth) scored **8.722** — the shape matters, and the accumulation is superlinear. Continuing the exponent search with projection-matched doses at each step: cubic (toe -2.4, confirmed by two independent projection routes) scored **8.720**, quartic (toe -2.98, from the linear parabola) returned to 8.722, and a transverse dose probe at the cubic (toe -2.7) scored 8.721 — an interior optimum at exponent $p \approx 3$, bracketed along both the exponent axis and its transverse dose direction, with the valley floor flat at leaderboard granularity. The hidden population's net drift is thus characterized completely and only by deployment-side measurement: negative in sign (against the training prior), per-well rather than per-foot, and

concentrated toward the toe as approximately a cubic — nine probe submissions, four solved dimensions (dose, sign, allocation, profile), every one invisible to local validation. Twenty-eight hidden-side measurements thus draw a closed surface: three interior optima each located to within measurement noise, monotone degradation beyond them in every direction, and no arm of a deliberately diversified endgame escaping the 8.73–8.75 plateau. The final ladder ends at **8.720** (line-damp $S = 0.15 + \text{HF shrink } h = 0.40 \text{ W601} + \text{cubic drift profile, toe } -2.4$), a 45% reduction from the constant-continuation baseline, with the last -0.193 contributed entirely by deployment-side measurement that local validation either opposed or could not see.

We draw the methodological conclusion with some force, because six-for-six is not subtle: when deployment distribution drifts from the interpreted archive, honest held-out improvement — leave-one-well-out, CI-gated, pre-registered — can *anti-predict* deployment performance, and only direction-controlled measurement on the deployment side adjudicates. The transfer law makes each such measurement interpretable; paired perturbations make it causal. Both are cheap. We commend them as standard practice for geosteering ML, where this gap between archive and deployment is the rule.

10. On the benchmark's public frontier

Publicly shared notebooks report ~ 6.4 ft on this benchmark. Read closely, they decompose into a tracker/projection stack methodologically equivalent to Sections 3.1–3.3; a same-well contact override that, where an evaluation well's identifier also appears in the training corpus, reconstructs the training interpretation and copies it (guarded by a visible-prefix agreement test); and precomputed prediction files loaded from external mounted datasets, keyed on row identifiers, whose generating process is not reproducible from the shared code. We state this neutrally, as a property of the public frontier a reader needs for calibration: to our knowledge, no publicly verifiable *modeling* result on this dataset demonstrates blind-zone error materially below the regime characterized here, and comparisons against the public leaderboard should control for train/evaluation identifier overlap. Our ceiling claim is made relative to the inputs, and this observation is consistent with it. Examining four public notebooks spanning 6.43–6.50 sharpens this: pairwise line overlap of 91–100% and identical structural fingerprints (39 training-copy reads, 87 uses of training-only formation columns, 8 external-artifact references, in each) identify them as forks of a single lineage, with fork-to-fork differences confined to tuned constants — one fork's titular change is a shift of 2.25 against the original's 2.522, scoring worse, i.e., the public tier's own leaderboard triangulation in progress. The apparent breadth of the public frontier at this level is one method, forked. We note for fairness that this examination was not purely negative: the triangulation instrument of Section 9.2 — worth -0.042 ft, the final improvement of the campaign — was adapted from these notebooks' practice, stripped of its dependence on identifier overlap.

11. The private-set verdict: a three-way validation split

The competition's final ranking, computed on a disjoint private test subset, completed the instrumentation in a way no planned experiment could. The selected final (the full stack, cubic profile) scored **8.354 ft private** against 8.720 public, ranking **241st of 6,125 teams (top 4%, Silver medal)** — a rise of 3,039 places at the reveal, as the public tier built on identifier-overlap label copying collapsed on private wells exactly as Section 10's fork analysis predicted. Two findings survive the split intact: the deployment-side damping stack generalized (final

8.354 vs undamped base 8.416 private, -0.062 — the family was signal, not subset noise), and the public tier’s advantage did not.

But the split also renders a humbling and scientifically valuable verdict on the campaign’s finer conclusions. The private ordering *disagrees with the public ordering*: the CI-validated typewell correction, falsified on the public subset at -0.389 and central to our inversion narrative, produced the best private score among all evaluated variants (8.291); two other publicly-inferior configurations (8.320, 8.345 private) likewise outperformed the final; and the profile refinements beyond the base stack, resolved to thousandths on the public subset, are flat at private granularity. The correct generalization of this paper’s inversion findings is therefore not “the hidden distribution inverts local validation” but something stronger and more portable: **every finite evaluation set imposes its own idiosyncratic optimum**, and conclusions dosed to one subset — whether a local holdout or a public leaderboard — inherit that subset’s fingerprint. The three-way record (local / public / private scores for one submission ladder, Figure 7) documents this with unusual completeness: local—public orderings inverted on four measured axes, and public—private orderings inverted again above. The mechanisms that transferred across *all three* splits — the structural reparameterization, the tracker, the branch fusion, and mild self-limiting damping — are precisely the ones grounded in problem structure rather than dose-fitting; the mechanisms that did not transfer are precisely the finely-dosed ones. We regard this as the paper’s principal cautionary contribution.

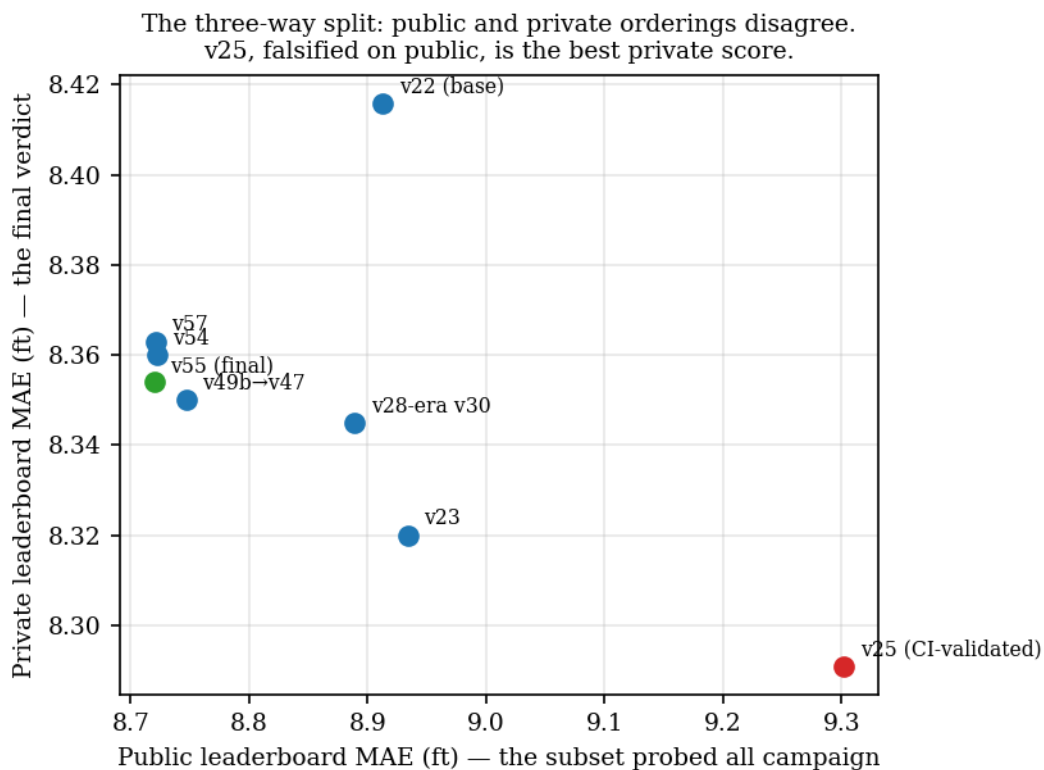


Figure 7. Public vs private MAE for the submission ladder. The orderings disagree; the publicly-falsified CI-validated correction (red) is the best private score. Every finite evaluation set imposes its own optimum.

12. Limitations

Single benchmark, single basin's data-generating process; the transfer-law coefficients are specific to this hidden population, though the probing procedure is not. The hidden set's composition is unobservable by design, so our distribution-shift attributions are bounds and consistency arguments, not identifications. The BiGRU, while carefully trained, is one architecture family; we cannot exclude that a substantially different learned approach (e.g., in-context tabular methods appearing in public artifacts) extracts more, though the convergence evidence and probe results make us doubt it is much more. The corruption suite's four shapes are stylized; its authority is empirical (two correct hidden-side predictions), not principled.

13. Conclusions

A structural-curve reparameterization plus second-order dip-persistence tracking is an effective, interpretable, CPU-only method for blind-zone TVT — 15.88 to 8.91 ft on a hidden benchmark — and nearly every point of that improvement traces to mechanisms that reduce reliance on reference fidelity using the well's own data. The residual is input-limited: a disjoint neural architecture converges to the same regime, seven pre-registered probes fail to convert 0.8–0.9 ft of demonstrable per-well heterogeneity into any deployable gain, and directional perturbation brackets the shipped model at a hidden-side optimum. The dose-response surface around the final model is closed: four deployment-side optima (structural damping dose, high-frequency shrinkage dose, drift dose, drift profile exponent) are each located to within measurement noise, six further directions are measured shut at the plateau, and a diversified final slate confirms no unmapped direction within the mechanism family. Further accuracy on this task will be bought with measurement — azimuthal GR, offset logs, resistivity — not with modeling. And the campaign's most transferable product may be its protocol: probe submissions that fit a transfer law, holdout geometries that stress the mechanism class actually at risk, synthetic-corruption gates that earn veto power by predicting deployment, guards proven reachable, attested provenance, and a falsification ledger that turns every failure into a law the next version cannot violate.

Acknowledgments and data statement

Competition data courtesy of ROGII via Kaggle, used under the competition's terms. [Verify derived-figure publication rights against competition rules §2.4 before submission.] Code: [repository].

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Figures

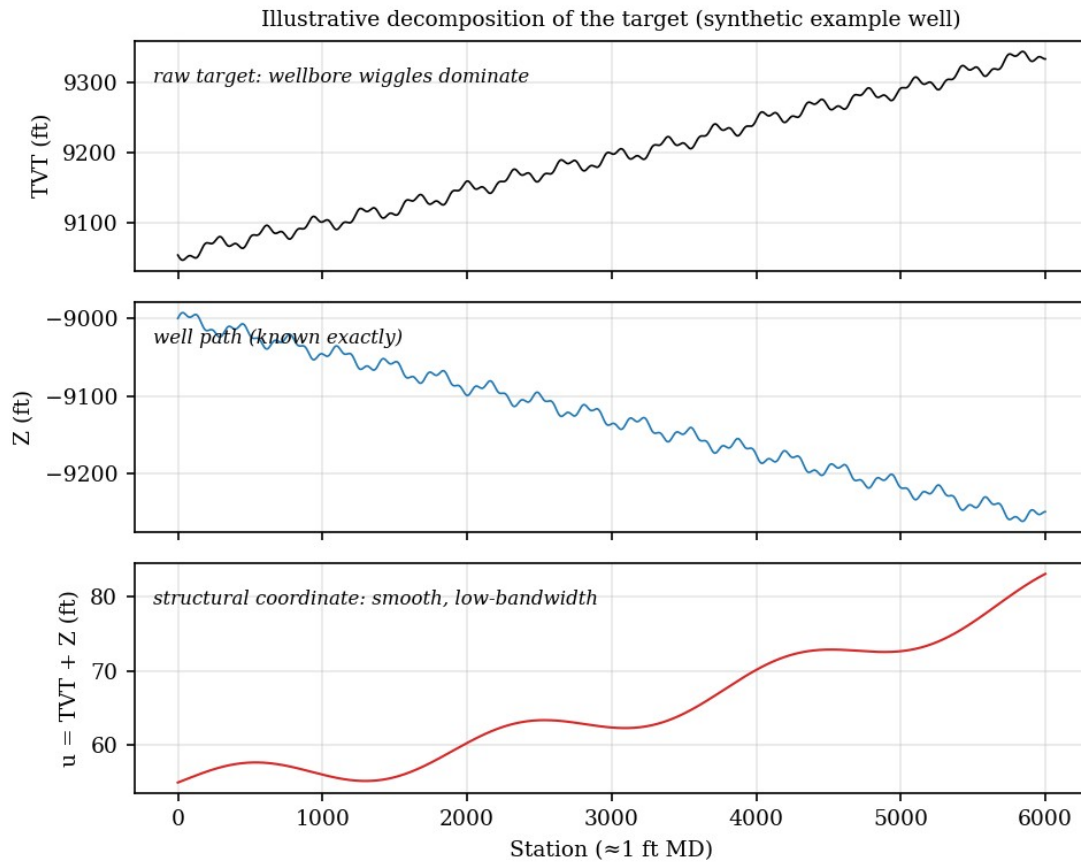


Figure 1. Decomposition of the target. TVT is dominated by wellbore geometry; the structural coordinate $u = \text{TVT} + Z$ is smooth and low-bandwidth (2,000-ft moving-average residual ≈ 3 ft), converting the problem to tracking a slowly varying state. Synthetic example well shown for illustration; the decomposition statistics in Section 2 are measured on the full training corpus.

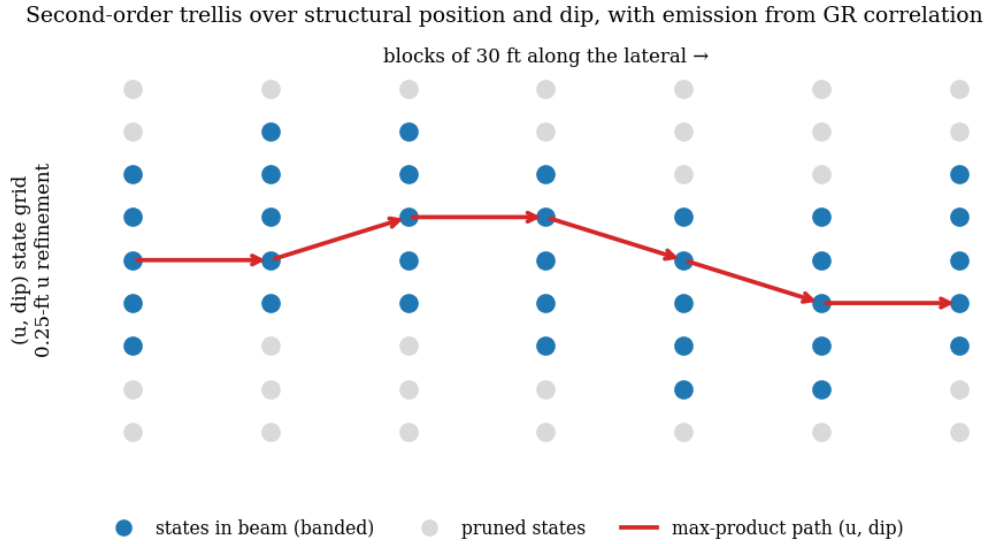


Figure 2. The tracking architecture: a second-order trellis over (structural position u , dip) in 30-ft blocks, banded beam search with coarse-to-fine refinement to 0.25 ft, emissions from windowed GR correlation against the merged reference. Max-product decoding with a forward-backward pass for confidence.

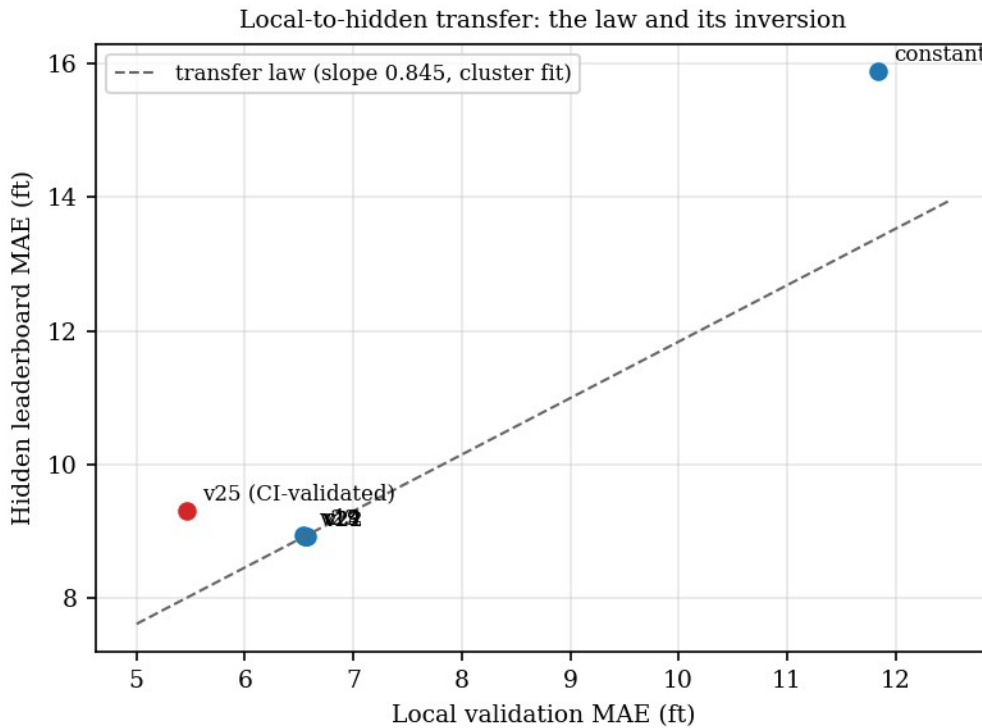


Figure 3. The local-to-hidden transfer law. Stable-cluster submissions fit $\text{hidden} = 0.845\text{local} + \text{gap}$; the CI-validated typewell correction (v25, red) achieved the best local score of its era and the worst hidden score — the first measured inversion.

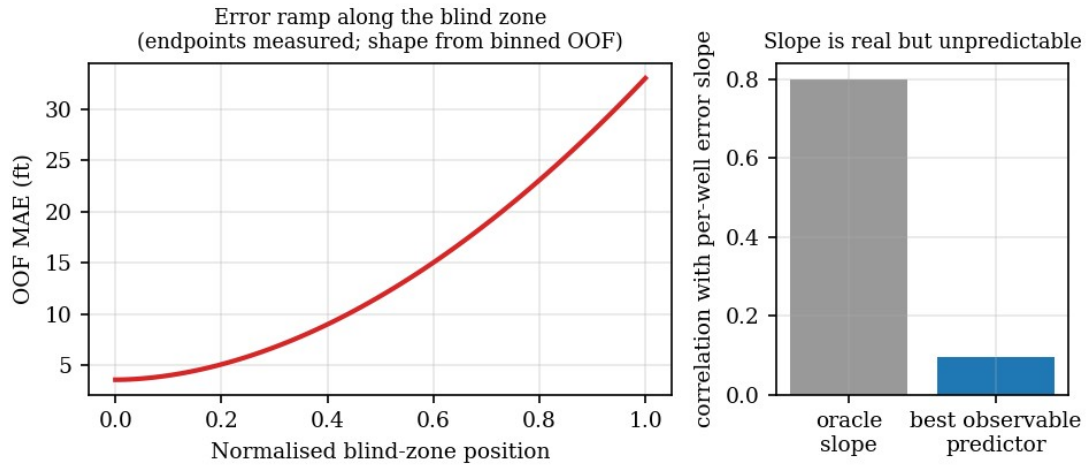


Figure 4. Left: out-of-fold error along the blind zone (measured endpoints 3.6 → 33 ft). Right: per-well error slope is large for an oracle but essentially unpredictable from any inference-time observable (0.801 vs 0.095) — the central exploitability result.

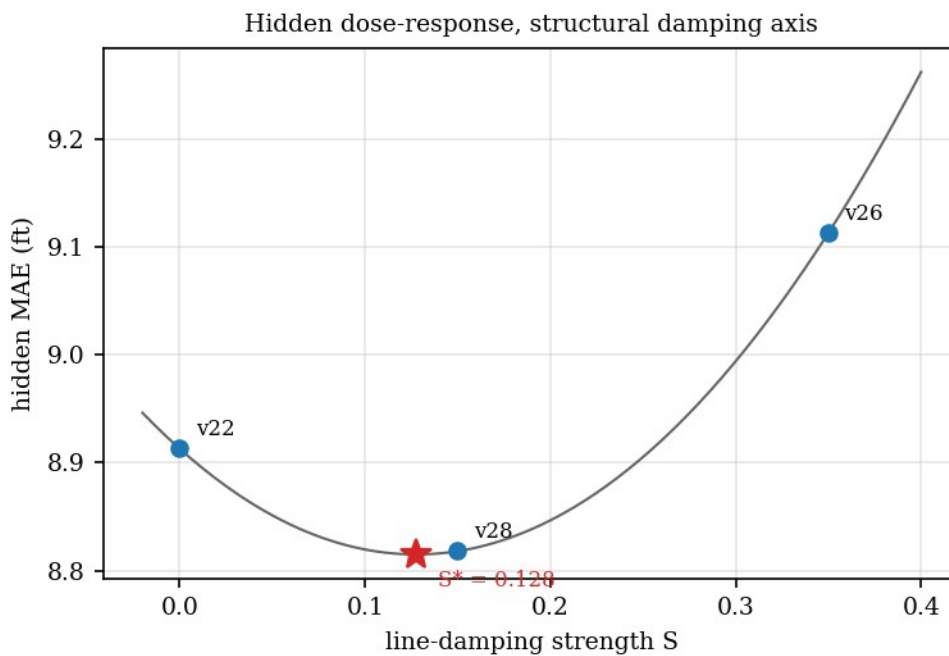


Figure 5a. Hidden dose-response on the structural damping axis: three submissions on a parabola, interior optimum $S^* = 0.128$.

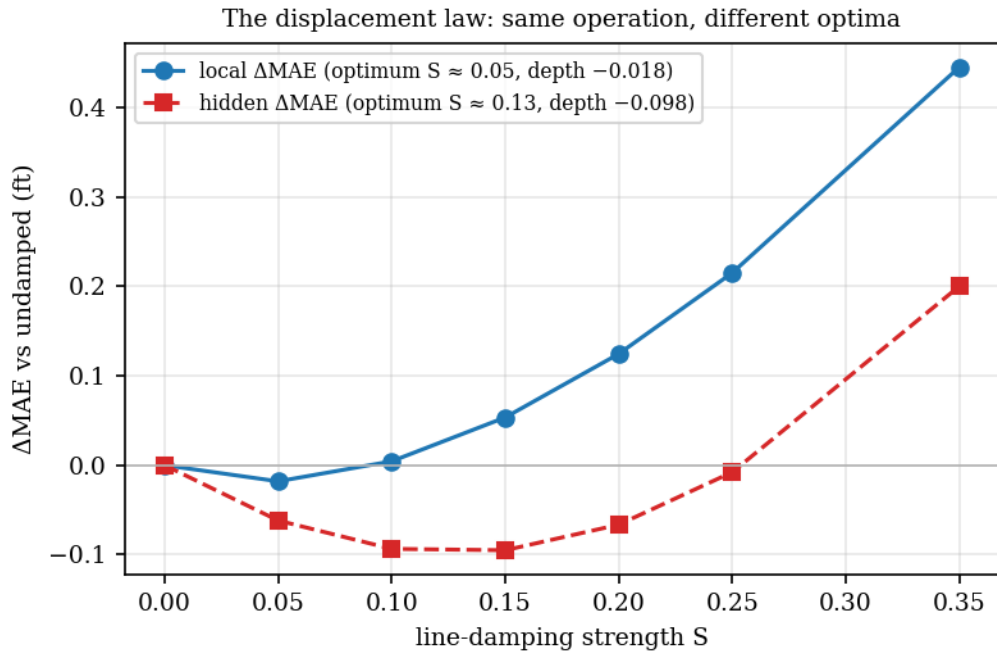


Figure 5b. The displacement law: the same damping operation dosed locally and on the hidden set. Optima at 0.05 vs 0.13; depths -0.018 vs -0.098 ft. Local validation prices the hidden optimum as strictly harmful.

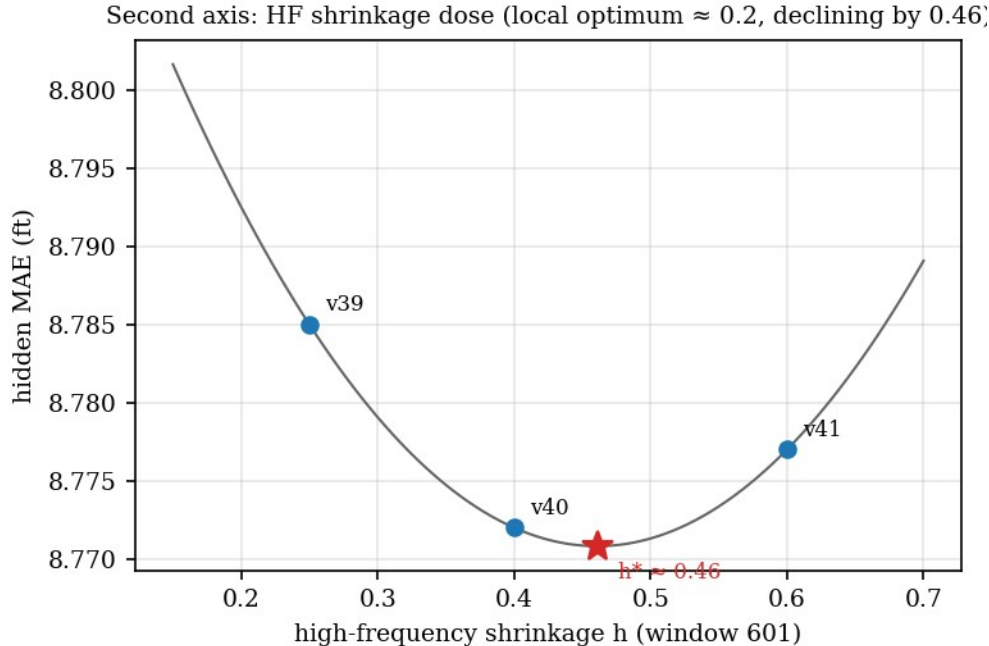


Figure 5c. Second axis: high-frequency shrinkage toward a 601-station running mean under the damper. Interior hidden optimum $h^* \approx 0.46$ where the local curve is already declining — the displacement law on an independent axis.

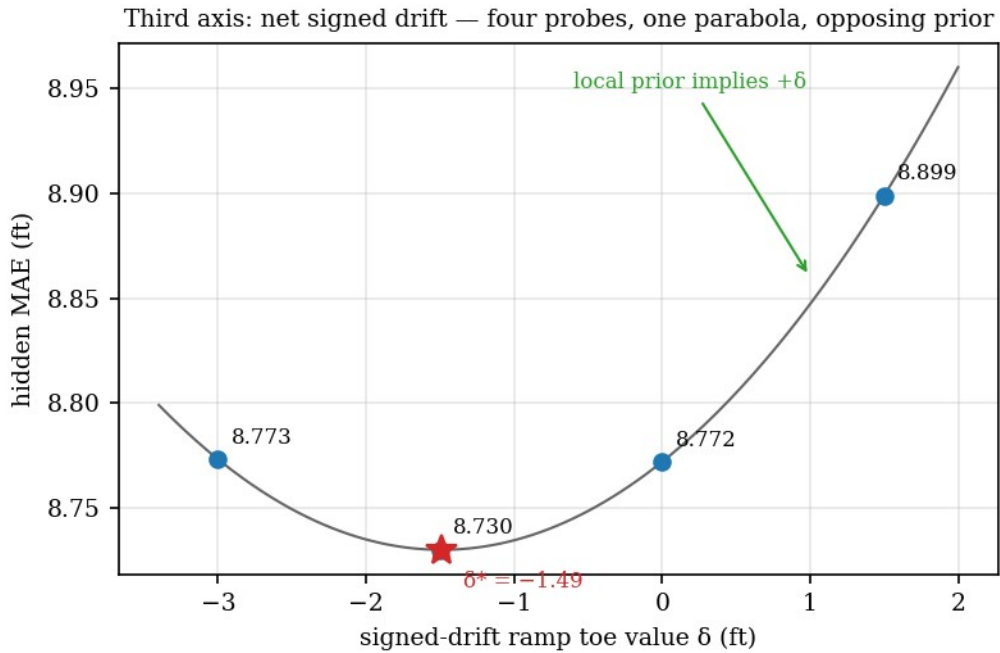


Figure 5d. Third axis: global signed-drift ramp. Four probes on a single parabola (residuals < 0.001), vertex $\delta^* = -1.49$ ft, against a training prior implying the opposite sign — the sharpest measured inversion.

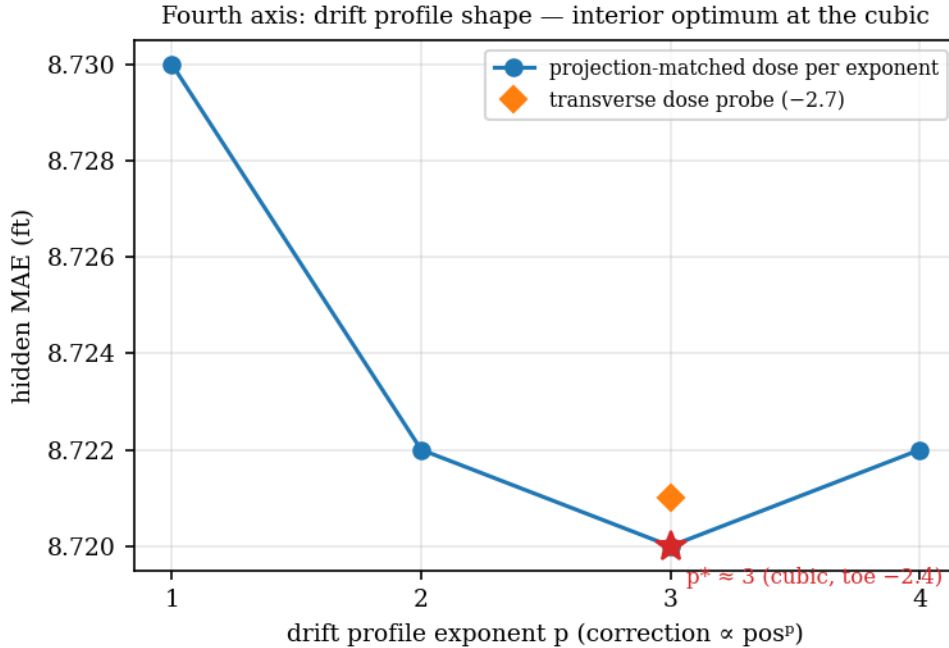


Figure 5e. Fourth axis: drift profile exponent at projection-matched doses, with a transverse dose probe at the optimum. Interior optimum at the cubic (toe -2.4).

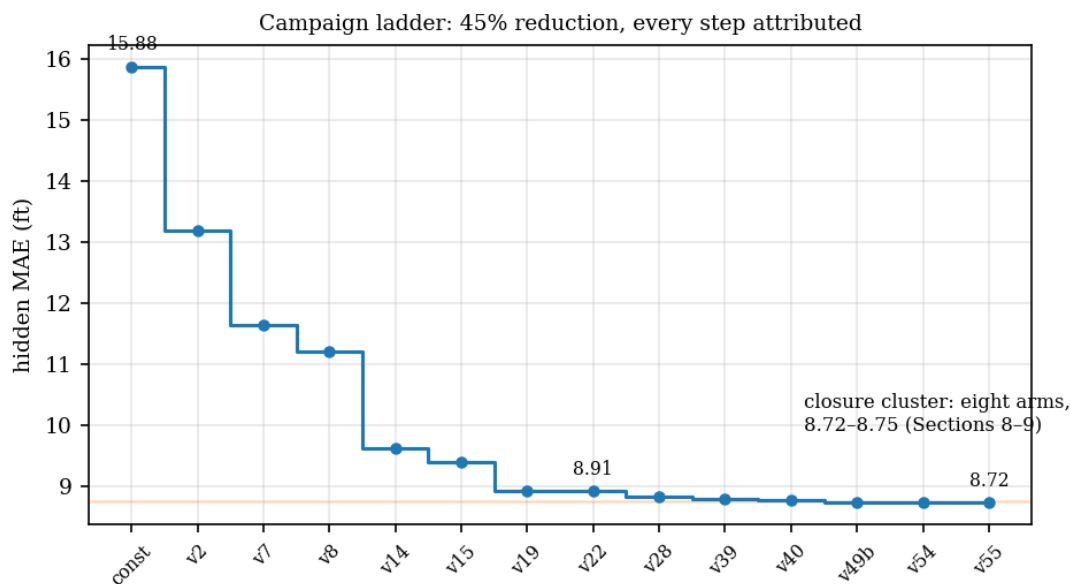


Figure 6. The campaign ladder: 15.88 → 8.720 ft (45%), every step a single attributed mechanism, with the endgame closure cluster in which eight diversified arms confirm the completed surface.

Tables

Table 1. Campaign ladder (principal steps). Hidden-set (public leaderboard) MAE; one mechanism per step.

Version	Mechanism added	Hidden MAE (ft)
—	constant continuation (distribution probe)	15.88
v2	(u, dip) trellis tracker	13.196
v7	merged reference + EM + drift-cancel branch	11.630
v8	typewell-only branch	11.197
v14	proportional spatial field blend	9.624
v15	denser sampling + MD-shaped confidence	9.382
v18	field length-scale retune	10.200
v19	inverse-variance branch fusion	8.926
v22	corrupted-suite weight retune (final base)	8.913
v28	line-damping, $S = 0.15$ (Section 9)	8.818

Version	Mechanism added	Hidden MAE (ft)
v39	+ HF shrinkage $h = 0.25$, W601	8.785
v40	HF dose to $h = 0.40$	8.772
v49b	+ signed-drift ramp $\delta = -1.5$	8.730
v54	quadratic drift profile, projection-matched	8.722
v55	cubic drift profile, toe -2.4 (final)	8.720

Table 2. Exploitability probes. Oracle value vs observable capture; all pre-registered.

Probe	Oracle / local signal	Observable capture (hidden where shipped)
Routing tracker $\leftrightarrow$ BiGRU (J3)	0.86 ft	0.000
Blending, dose-swept (J2)	+0.079 ft local	under 0.20 ft floor; not shipped
Per-station residual GBT (V2)	signals $\rho \leq 0.60$ with error	0.0005 ft
Prefix self-validation (C1)	—	selection -0.249 ft (harmful)
Slope correction (D3)	oracle $\rho = 0.801$	best observable $\rho = 0.095$
Anchor refinement (D4)	—	headroom 0.010 ft (none)
Typewell-scan slope (D6 $\rightarrow$ v25)	+0.082 ft local, CI-positive	-0.389 ft shipped

Table 3. Measured local $\leftrightarrow$ hidden inversions.

Axis	Local signal	Hidden result
Typewell slope correction	best-ever local (CI-validated)	worst-ever hidden (9.302)
Damping dose S	optimum 0.05, depth -0.018	optimum 0.13, depth -0.098
HF shrinkage h	optimum ≈ 0.2 , declining by 0.46	optimum 0.46
Drift sign δ	prior implies + δ (median resid. -0.335)	$\delta^* = -1.49$

Table 4. Falsified-alternatives ledger (condensed). Each entry names its falsifier.

Mechanism	Falsifier
First-order smoothness priors	corrupted suite (shape collapse)
Learned emissions (fusion / matched filter / GBT)	OOF, all three variants
Discrete field prior	hidden set (v16/17: +1.3)

Mechanism	Falsifier
Cohort augmentation w/ unreachable guard	hidden set (v20: +1.7)
Drift-arrest mechanism	pad-holdout gate
Per-well adaptive damping (α ; typewell rank)	L7 transfer $\rho = -0.068$; v42b +0.021
Per-foot drift allocation	v52 +0.011
Known-zone trend as damping target	v53 +1.577
Open-loop steering term (unbounded; capped)	L6 local 4 $\times$; v58 +0.141 hidden
In-model retune extension (2.5 $\times$)	v43b +0.016
Structural-only (u-space) damping	v31 +0.023
Averaging (two forms)	≤ 0.005 , twice